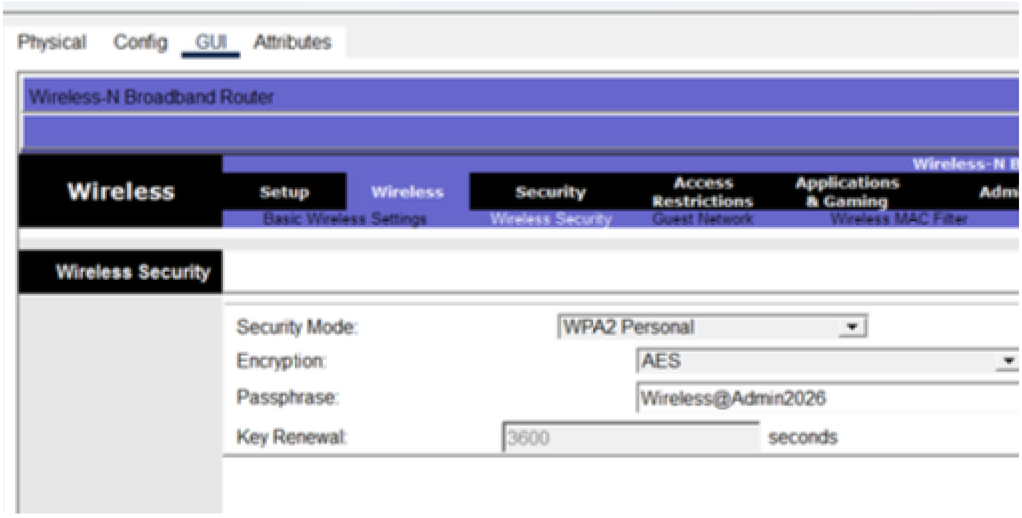
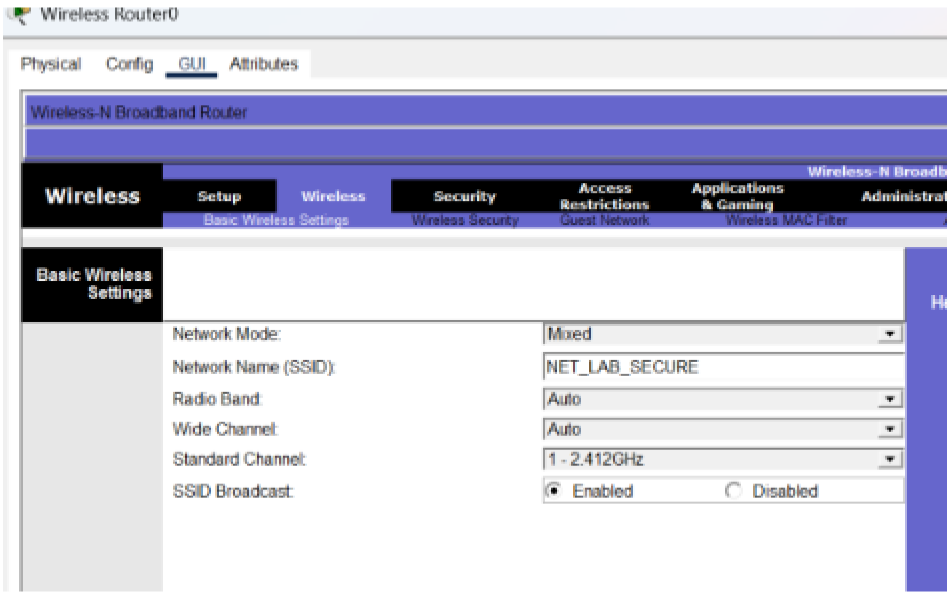
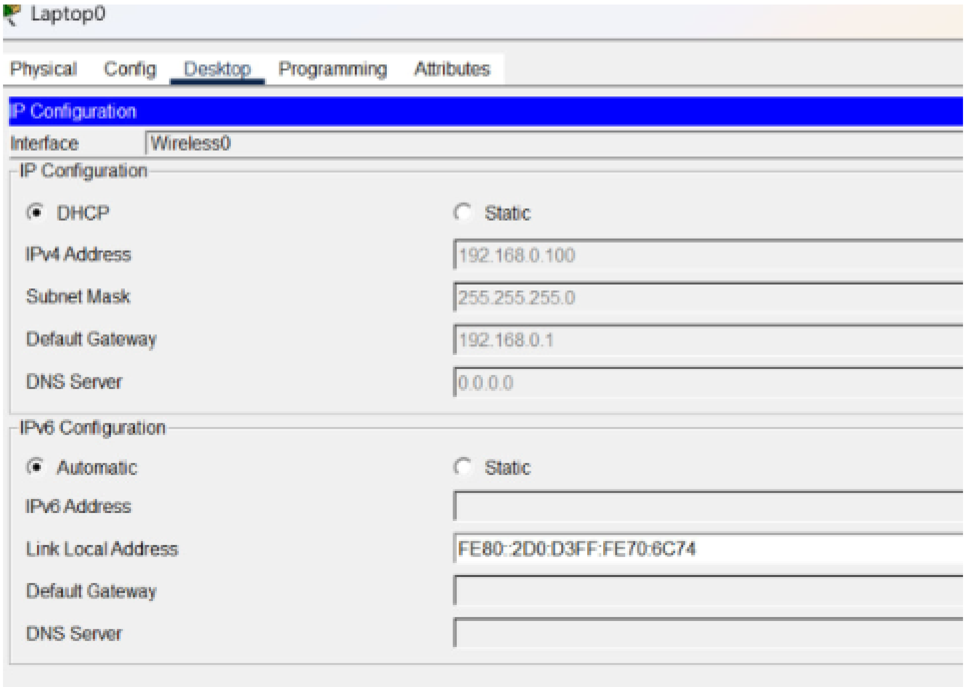
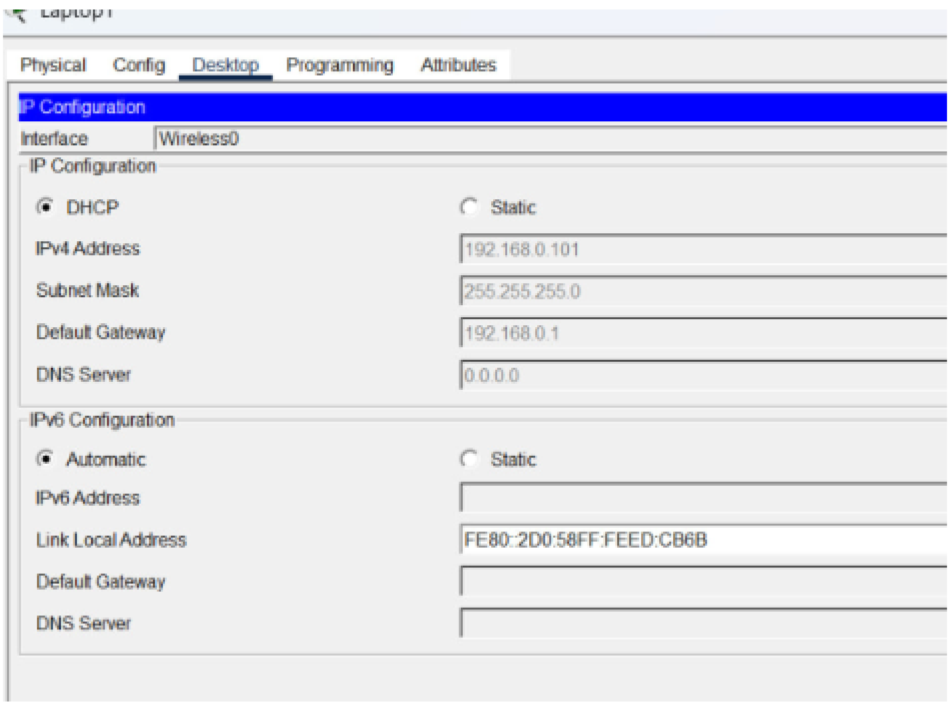
Lab 13
Course: Networks System Design
Name: Do Davin
Student ID: P20230018
Instructor: Mr. Kuy Movsun

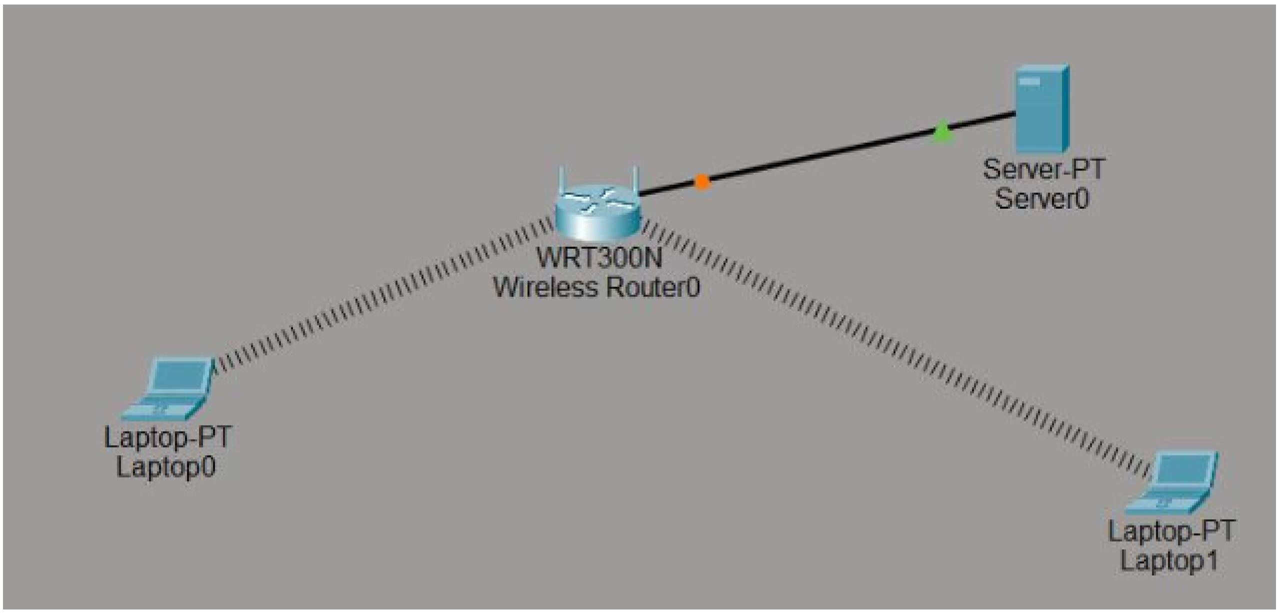
Part 1: Infrastructure Mode & Connectivity






Laptop0
Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
Wireless0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.100
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:D3FF:FE70:6C74
Default Gateway
DNS Server


Physical
Config
Desktop
Programming
Attributes
IP Configuration
Interface
Wireless0
IP Configuration
DHCP
Static
IPv4 Address
192.168.0.101
Subnet Mask
255.255.255.0
Default Gateway
192.168.0.1
DNS Server
0.0.0.0
IPv6 Configuration
Automatic
Static
IPv6 Address
Link Local Address
FE80::2D0:58FF:FEED:CB6B
Default Gateway
DNS Server




```
C:\>ping 192.168.0.100

Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=33ms TTL=128
Reply from 192.168.0.100: bytes=32 time=14ms TTL=128
Reply from 192.168.0.100: bytes=32 time=20ms TTL=128
Reply from 192.168.0.100: bytes=32 time=20ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 14ms, Maximum = 33ms, Average = 21ms
```

```
Pinging 192.168.0.100 with 32 bytes of data:

Reply from 192.168.0.100: bytes=32 time=3ms TTL=128
Reply from 192.168.0.100: bytes=32 time=4ms TTL=128
Reply from 192.168.0.100: bytes=32 time=3ms TTL=128
Reply from 192.168.0.100: bytes=32 time=4ms TTL=128

Ping statistics for 192.168.0.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 3ms, Maximum = 4ms, Average = 3ms

C:\>
```

Part 2: Analyzing Wireless Management Frames

Event List

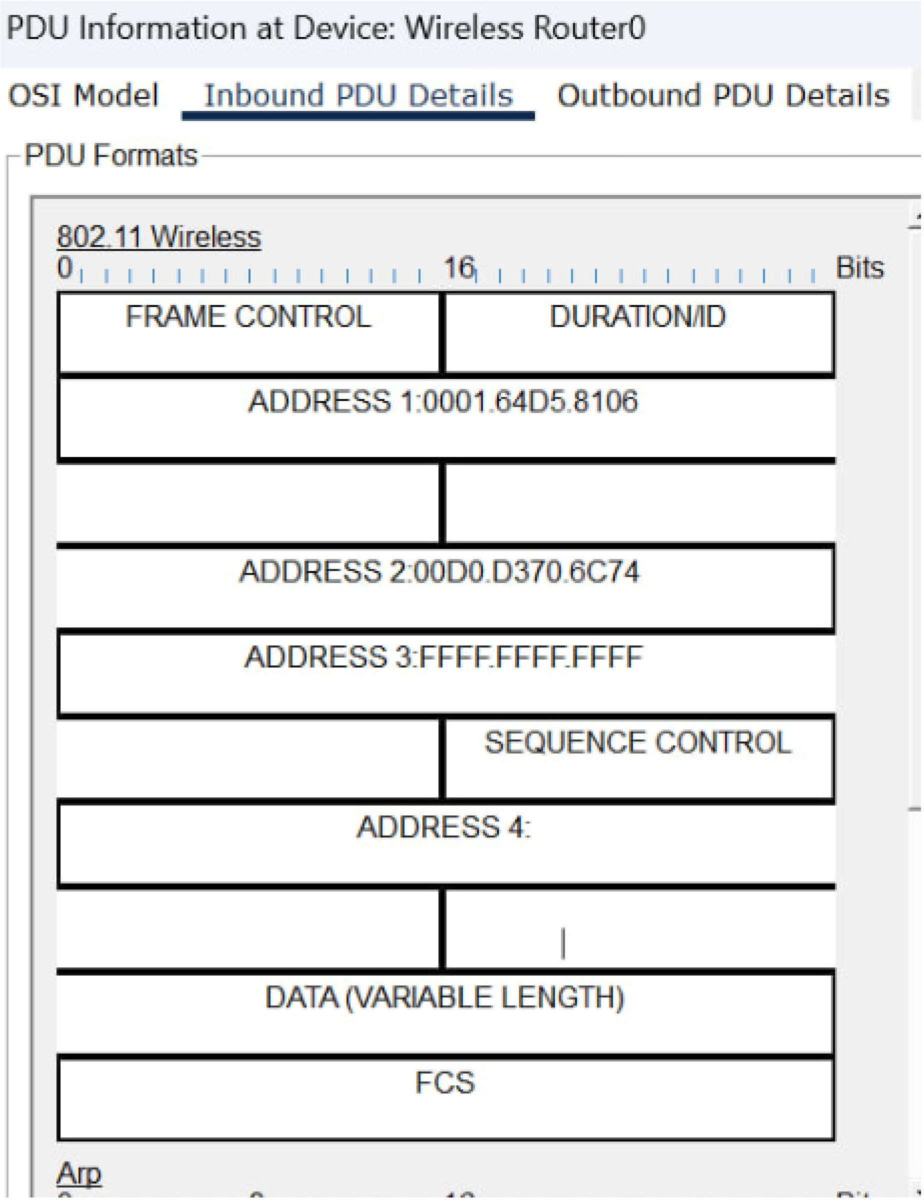
Vis.	Time(sec)	Last Device	At Device	Type
	0.000	--	Laptop0	ICMP
	0.000	--	Laptop0	ARP
	0.001	Laptop0	Wireless Router0	ARP
	0.002	Wireless Router0	Server0	ARP
	0.005	--	Wireless Router0	ARP
	0.006	Wireless Router0	Laptop1	ARP
	0.006	Wireless Router0	Laptop0	ARP
	0.011	--	Wireless Router0	ARP
	0.012	Wireless Router0	Laptop1	ARP
	0.012	Wireless Router0	Laptop0	ARP
	0.012	--	Laptop0	ICMP
	0.015	--	Laptop0	ICMP
	0.016	Laptop0	Wireless Router0	ICMP
	0.021	--	Wireless Router0	ICMP
	0.022	Wireless Router0	Laptop1	ICMP
	0.022	Wireless Router0	Laptop0	ICMP
	0.027	--	Wireless Router0	ICMP
	0.028	Wireless Router0	Laptop1	ICMP

PDU Information at Device: Wireless Router0

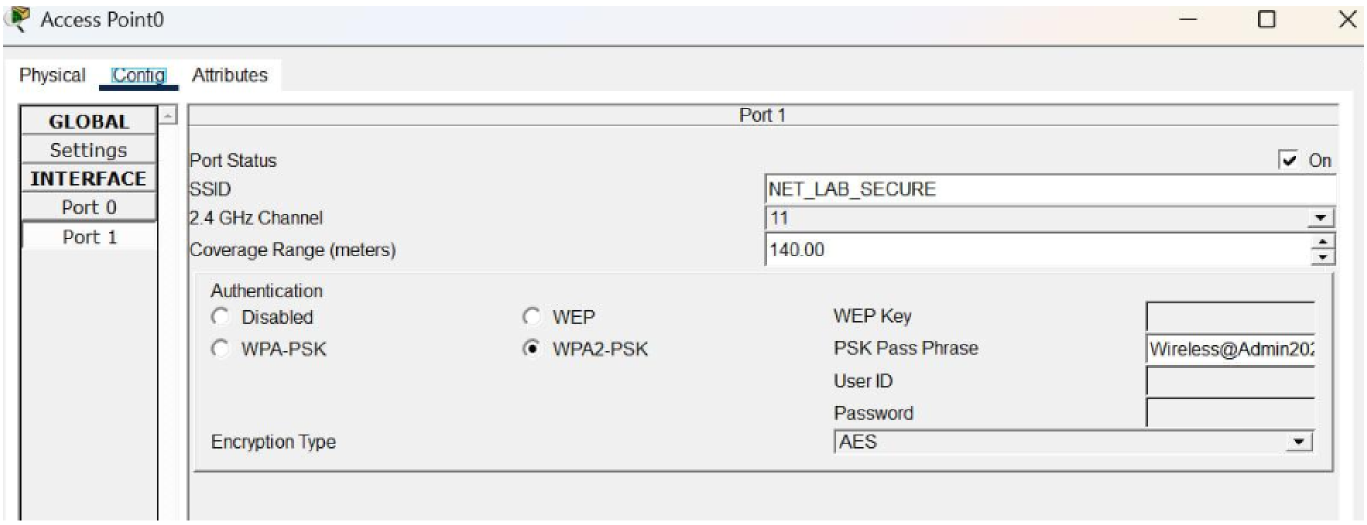
OSI Model | Inbound PDU Details | Outbound PDU Details

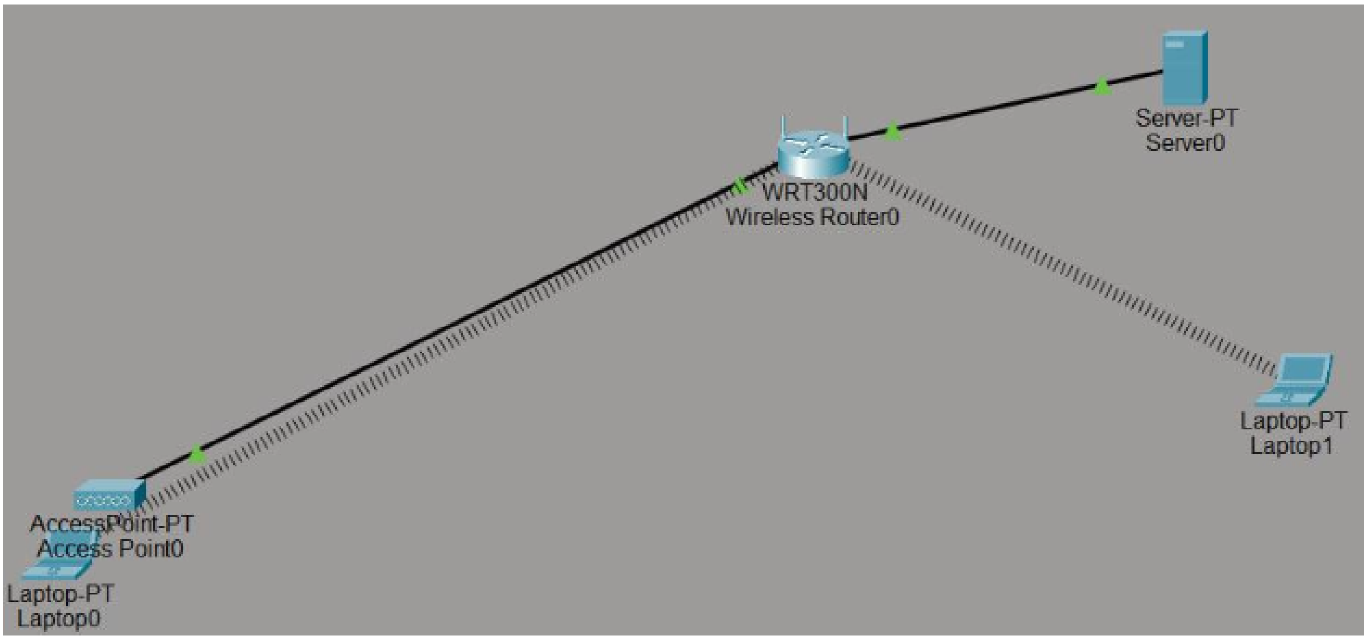
At Device: Wireless Router0
Source: Laptop0
Destination: Broadcast

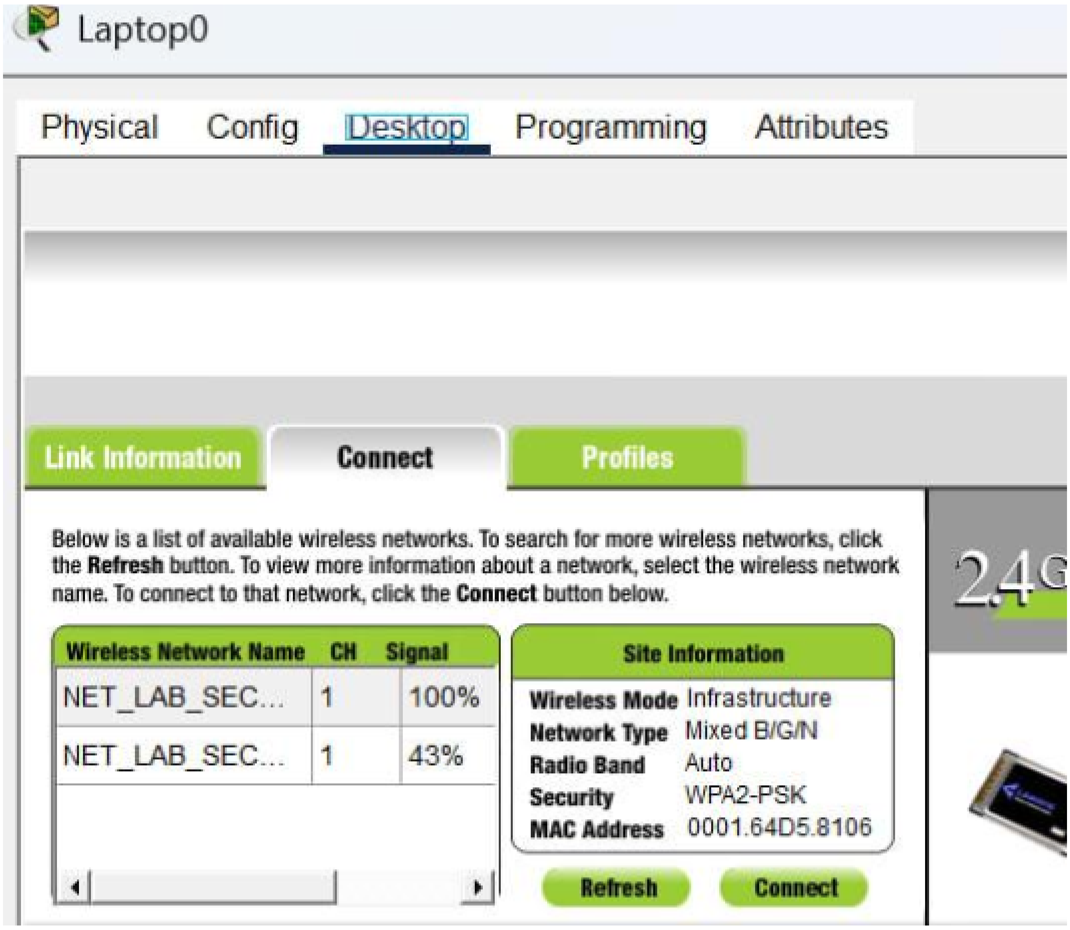
In Layers	Out Layers
Layer7	Layer7
Layer6	Layer6
Layer5	Layer5
Layer4	Layer4
Layer3	Layer3
Layer 2: Wireless ARP Packet Src. IP: 192.168.0.98, Dest. IP: 192.168.0.1	Layer 2: Ethernet II Header 00D0.D370.6C74 >> FFFF.FFFF.FFFF ARP Packet Src. IP: 192.168.0.98, Dest. IP: 192.168.0.1
Layer 1: Port Wireless	Layer 1: Port(s): Ethernet 1



Part 3: Channel Management and Frequency Reuse

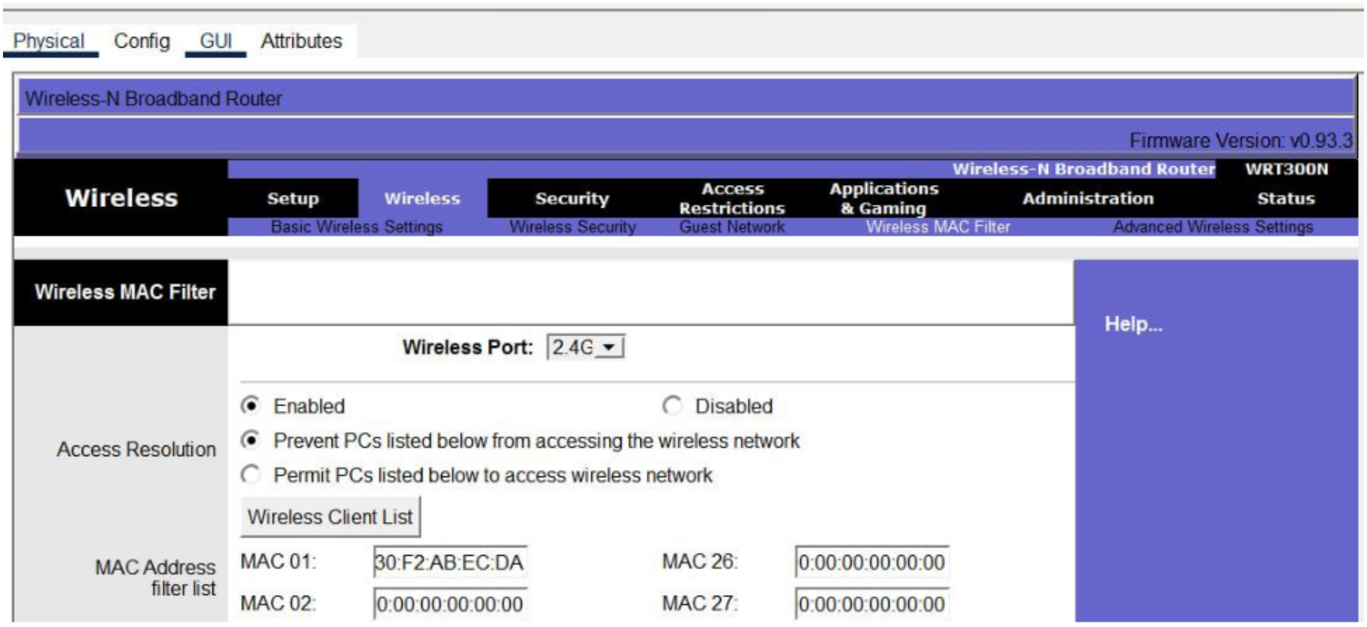


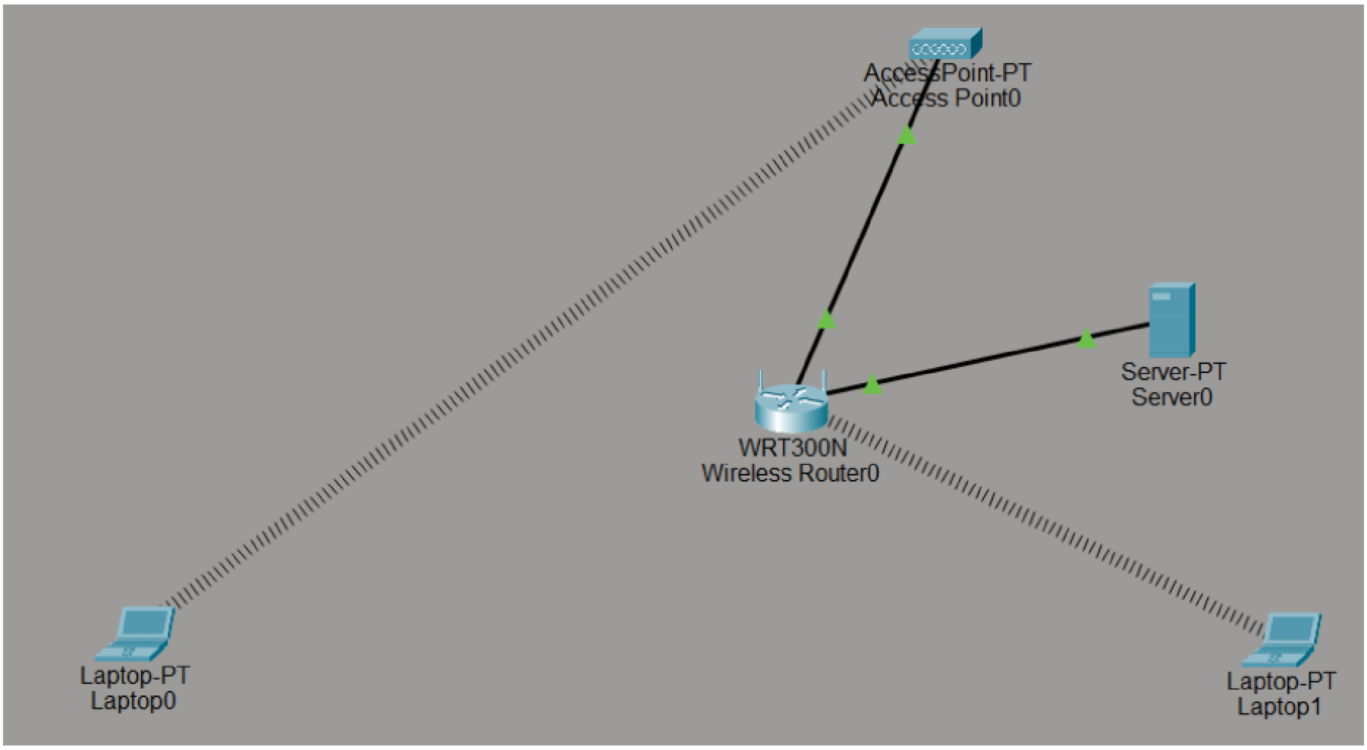




Part 4: Implementing Layer 2 Security

```
C:\>

C:\>ipconfig /all

Bluetooth Connection:(default port)

   Connection-specific DNS Suffix..:
   Physical Address................: 0030.F2AB.ECDA
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
   DHCP Servers....................: 0.0.0.0
   DHCPv6 IAID.....................:
   DHCPv6 Client DUID..............: 00-01-00-01-0E-56-64-39-00-D0-D3-70-6C-74
   DNS Servers.....................: ::
                                     0.0.0.0

Wireless0 Connection:

   Connection-specific DNS Suffix..:
   Physical Address................: 00D0.D370.6C74
   Link-local IPv6 Address.........: FE80::2D0:D3FF:FE70:6C74
 --More--
```





Questions:

1. Why do we use non-overlapping channels like 1, 6, and 11 in professional deployments?
2. What happened to the connection when you applied the MAC filter? Is this an effective security measure on its own?
3. In Simulation Mode, what is the main difference you noticed between a "Beacon" frame and an "Association" frame?

Answers:

1. We don’t use non-overlapping channels (1, 6, and 11) in professional deployments because it prevent interference between adjacent access points. 2.4GHz channels overlap, but these three have 25MHz separation so they don't interfere. Allows multiple APs in same area without performance degradation.
2. When you applied the MAC filter, the router blocked the connection from laptop0 immediately while Laptop1 worked normally. It’s not an effective security measure on its own because it easily bypass MAC spoofing.
3. **Beacon**: * AP -> Broadcast, periodic advertisements * ongoing advertisements **Association:** * Client -> Specific AP, connection request * one-time connection setup.